CS670: Cryptographic Techniques for Privacy Preservation

Module 4: Private Information Retrieval

Adithya Vadapalli

Drawings of Alice, Bob, and other crypto characters by Simon Oya: https://simonoya.com/drawings/

A few talking points

Assignments

Quiz 3 and 4 on November 4 and November 7

CS714 being offered next semester

UGPs and M.Tech/PhD Thesis

Course Feedback

Private Information Retrieval

Downloader

Wants to watch a movie on Netflix.

However, she does not want Netflix to know what she watched.

Private Information Retrieval

Can we download a record from a remote database server, without the server learning what we downloaded?

Yes! (under certain assumptions, though)

Private Information Retrieval

Private Information Retrieval is a cryptographic technique that allows a user to download from a remote database without the database owner learning what was downloaded!

In general, the user can learn more than what they asked for.

If we have an additional condition that the user is supposed to learn only the record they requested, we refer to it as Symmetric Private Information Retrieval (SPIR).

Private Information Retrieval

What is the difference between Private Information Retrieval and Anonymous Information Retrieval?

Tor is an example of anonymous information retrieval!

Some other examples where PIR is useful?

You may want to look up something in a patent database!

Private Information Retrieval

assumptions

Information Theoretic

Multiserver   No Computation Assumptions

Computational

Single Server   Computation Assumptions

Simplest PIR Protocol

What is the simplest PIR protocol?

Naively downloading the entire database!

Downloader

I want to watch a video

Here is the entire Netflix Catelogue

Is this SPIR?

Private Information Retrieval

Information Theoretic Private Information Retrieval (IT-PIR)

Multiple PIR Servers

Each holding the replica of database

(a matrix of 0s and 1s)

Goal: To retrieve the $\mathtt{j}^{\mathtt{th}}$ row (record) of the database (matrix) privately.

IT-Private Information Retrieval

(a matrix of 0s and 1s)

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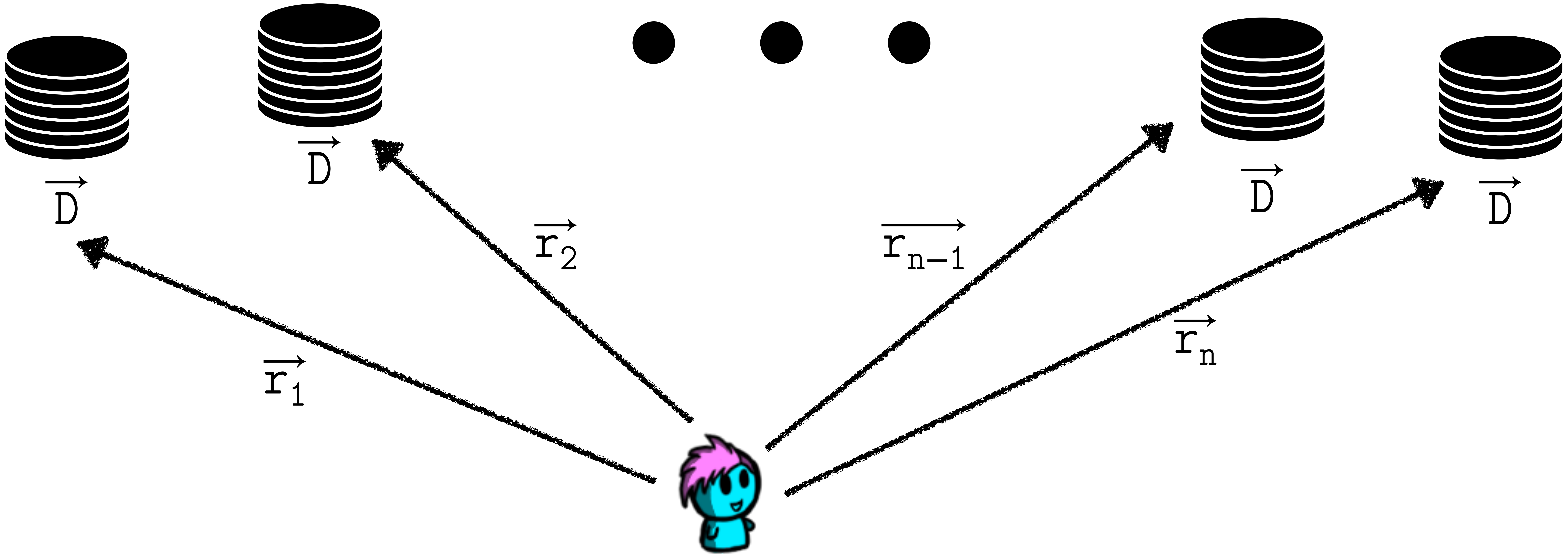

IT-Private Information Retrieval

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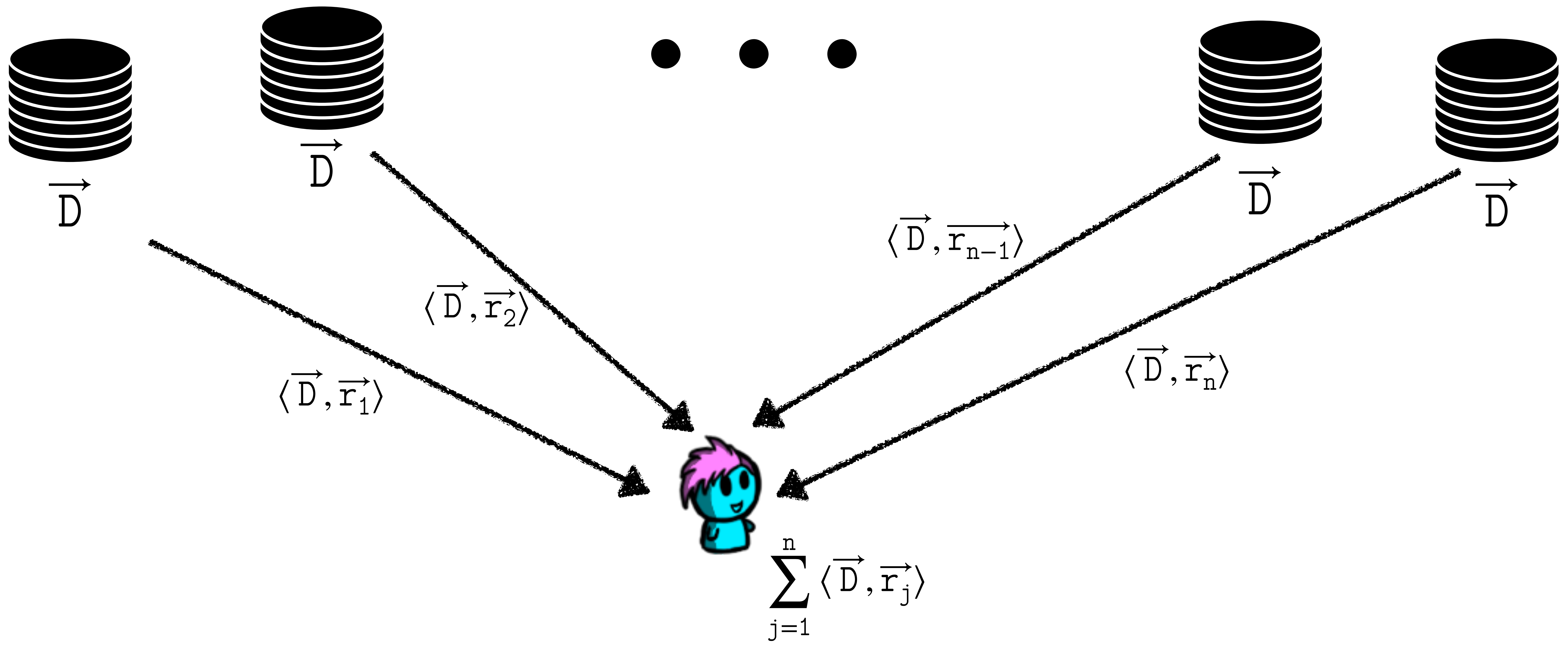
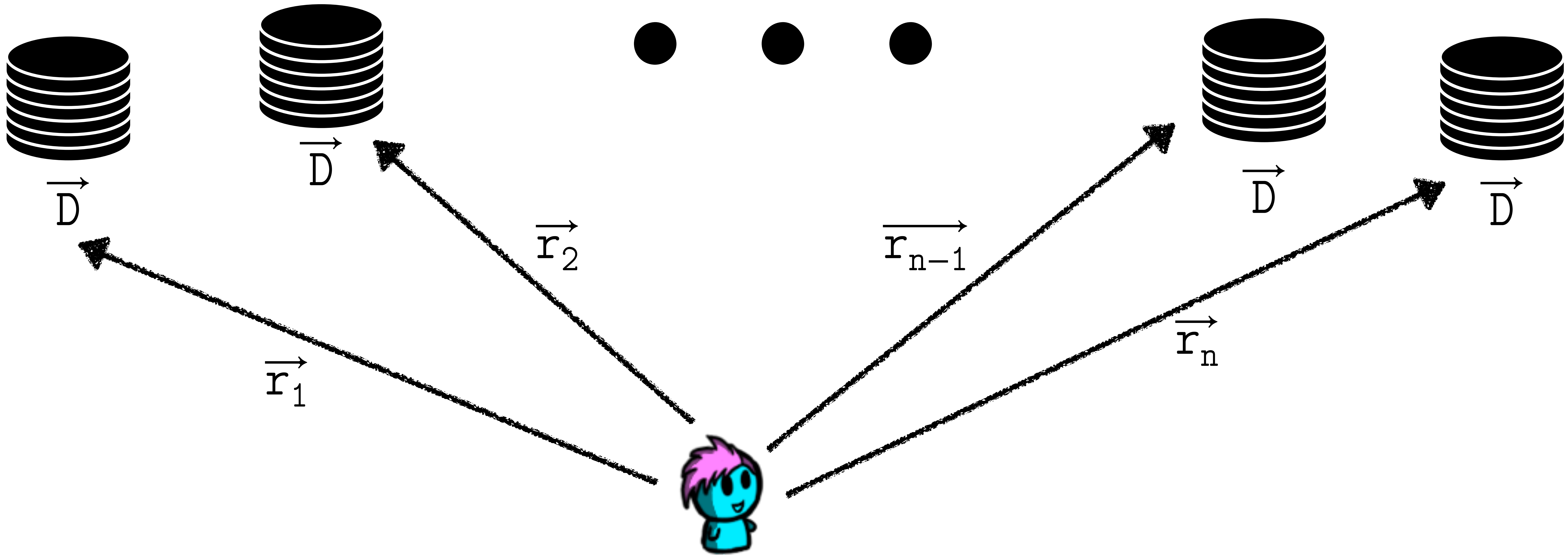


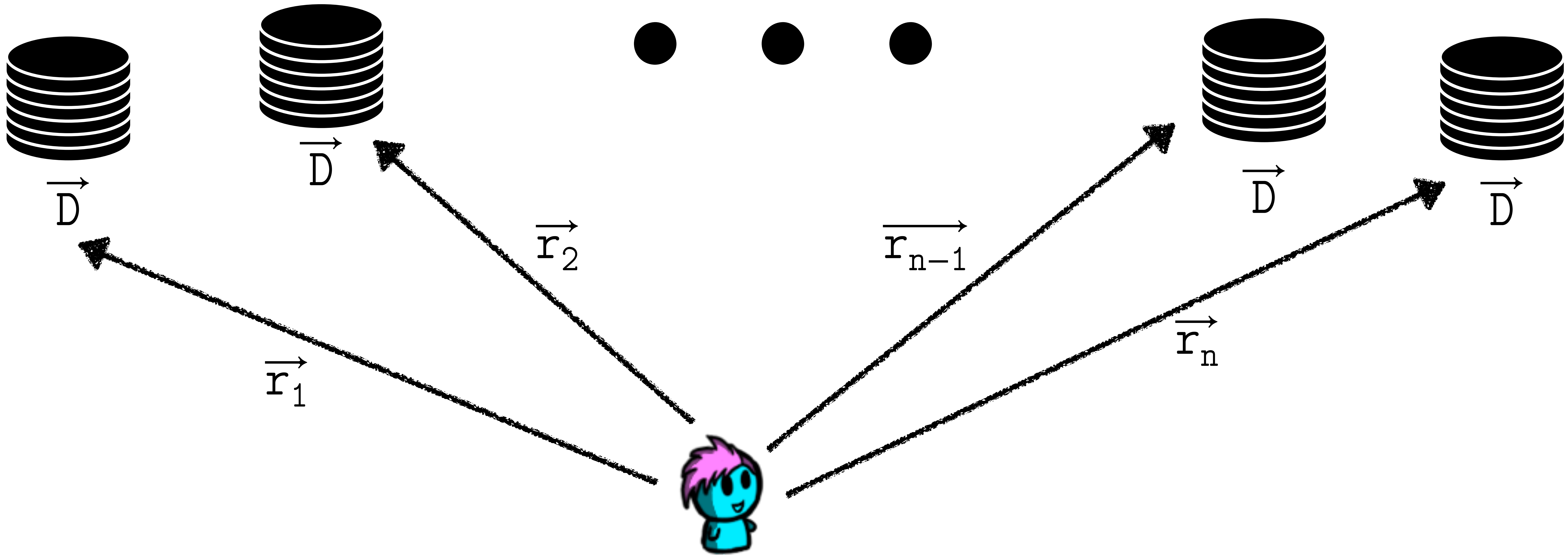
IT-Private Information Retrieval



Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

What should be the property of these vectors? $\sum_{j=1}^{n} \overrightarrow{r_j} = \overrightarrow{e_i}$

IT-Private Information Retrieval

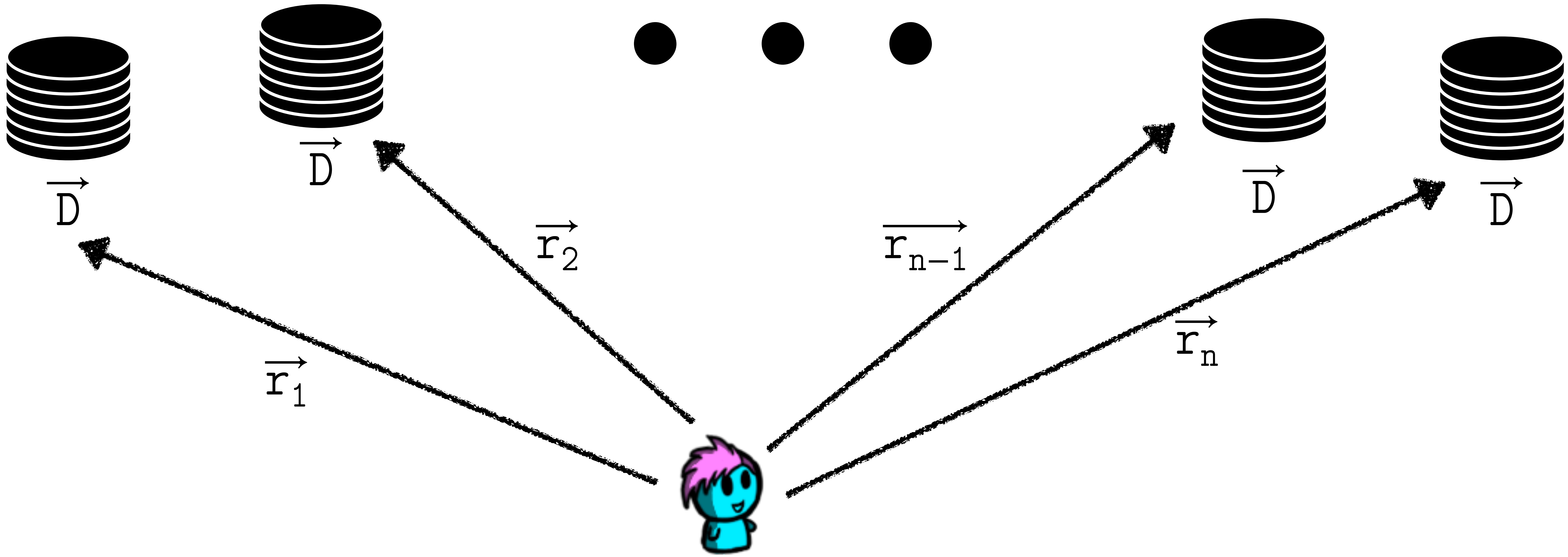


Goal: To retrieve the $\mathtt{j}^{\mathtt{th}}$ row (record) of the database (matrix) privately.

What is the upload cost?

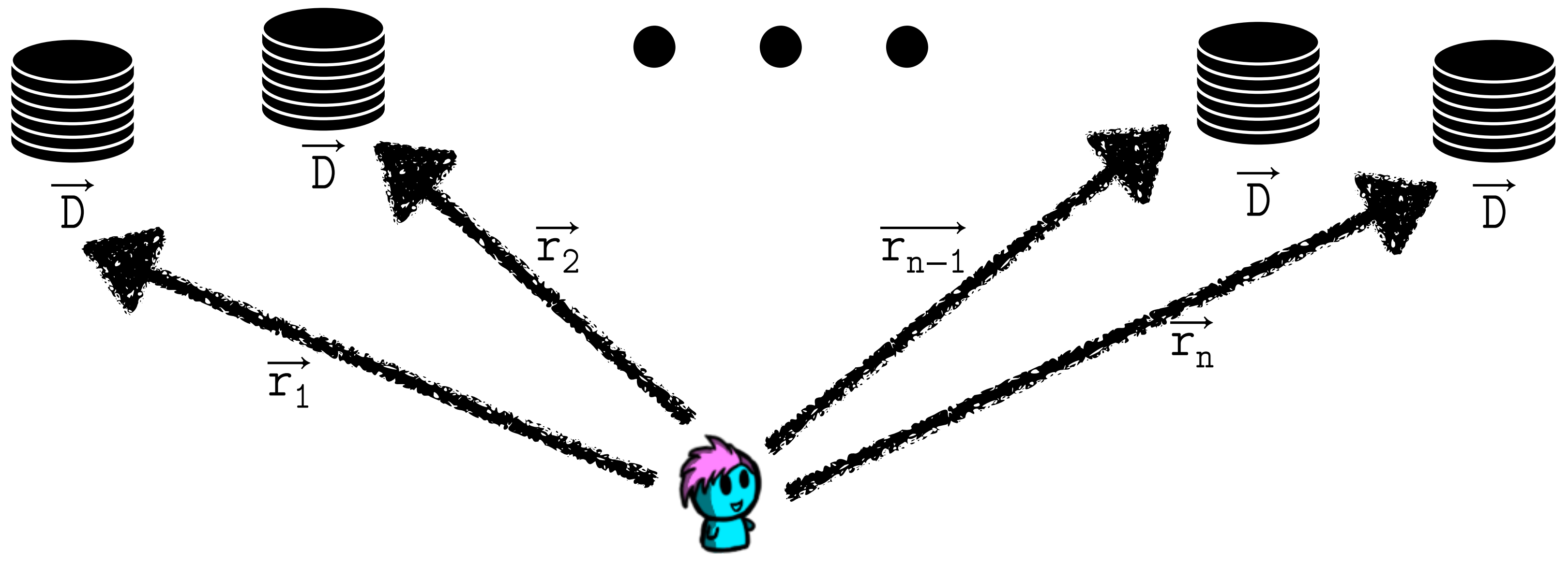
What is the download cost?

IT-Private Information Retrieval



What could go wrong?

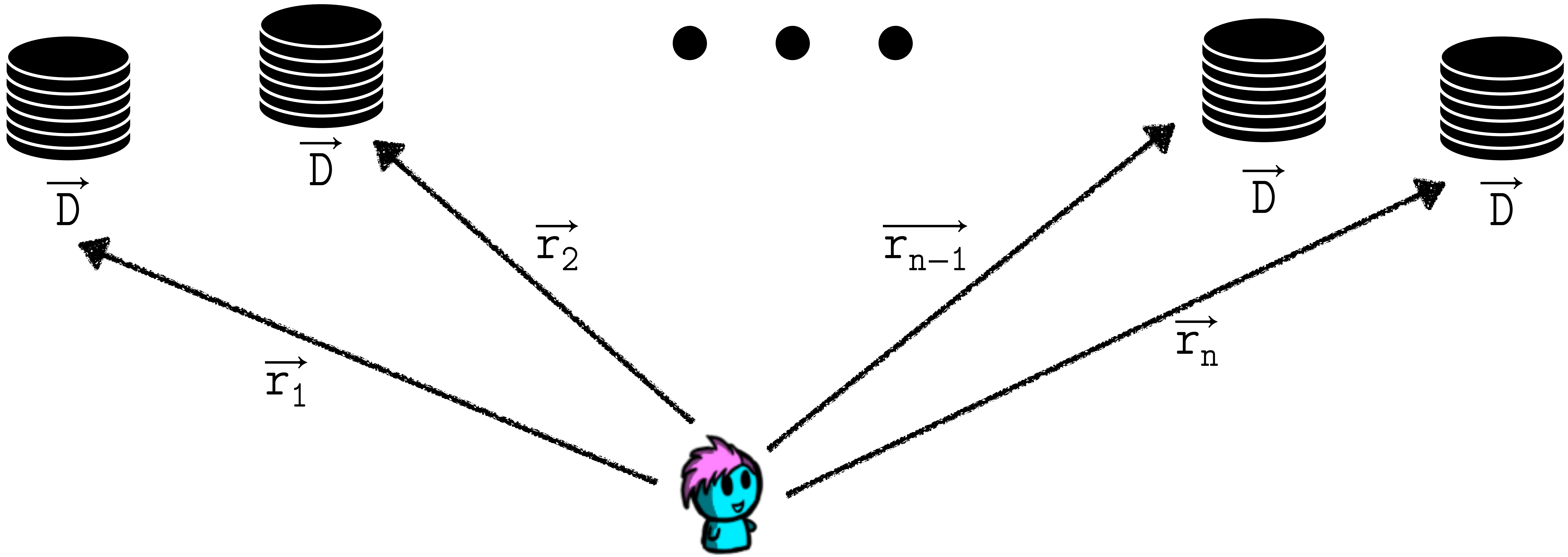
IT-Private Information Retrieval



How to protect against servers that do not respond?

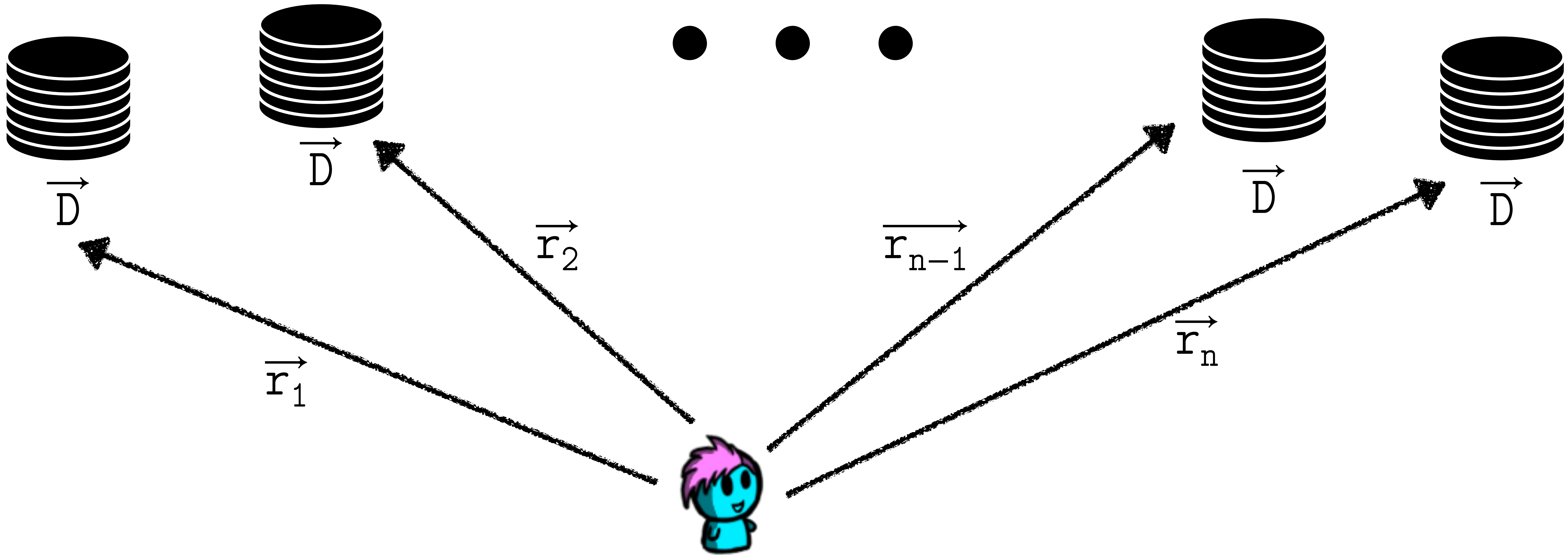
Perhaps we could use a different type of secret sharing?

IT-Private Information Retrieval



What are the downsides of this protocol?

IT-Private Information Retrieval



Can we use Distributed Point Functions here?

IT-Private Information Retrieval

$$\begin{array}{cccccc|c}
x_{00} & x_{10} & x_{20} & x_{30} & x_{40} & x_{50} & 0 \\
x_{01} & x_{11} & x_{21} & x_{31} & x_{41} & x_{51} & 0 \\
x_{02} & x_{12} & x_{22} & x_{32} & x_{42} & x_{52} & 0 \\
x_{03} & x_{13} & x_{23} & x_{33} & x_{43} & x_{53} & 0 \\
x_{04} & x_{14} & x_{24} & x_{34} & x_{44} & x_{54} & 0 \\
x_{05} & x_{15} & x_{25} & x_{35} & x_{45} & x_{55} & 0
\end{array}$$

$$q_1 = [0, 3, 0, 5, 3, 5]$$

$$\begin{array}{cccccc|c}
x_{00} & x_{10} & x_{20} & x_{30} & x_{40} & x_{50} & 0 \\
x_{01} & x_{11} & x_{21} & x_{31} & x_{41} & x_{51} & 0 \\
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x_{05} & x_{15} & x_{25} & x_{35} & x_{45} & x_{55} & 0
\end{array}$$

$$q_1 = [0, 3, 6, 5, 3, 5]$$

IT-Private Information Retrieval

$$
\begin{array}{ccccccc}
x_{00} & x_{10} & x_{20} & x_{30} & x_{40} & x_{50} & 0 \\
x_{01} & x_{11} & x_{21} & x_{31} & x_{41} & x_{51} & 0 \\
x_{02} & x_{12} & x_{22} & x_{32} & x_{42} & x_{52} & 0 \\
x_{03} & x_{13} & x_{23} & x_{33} & x_{43} & x_{53} & 0 \\
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x_{05} & x_{15} & x_{25} & x_{35} & x_{45} & x_{55} & 0
\end{array}
$$

$\vec{q} = [0 \ldots s]^r$ such that $\vec{q}[j] = 0$

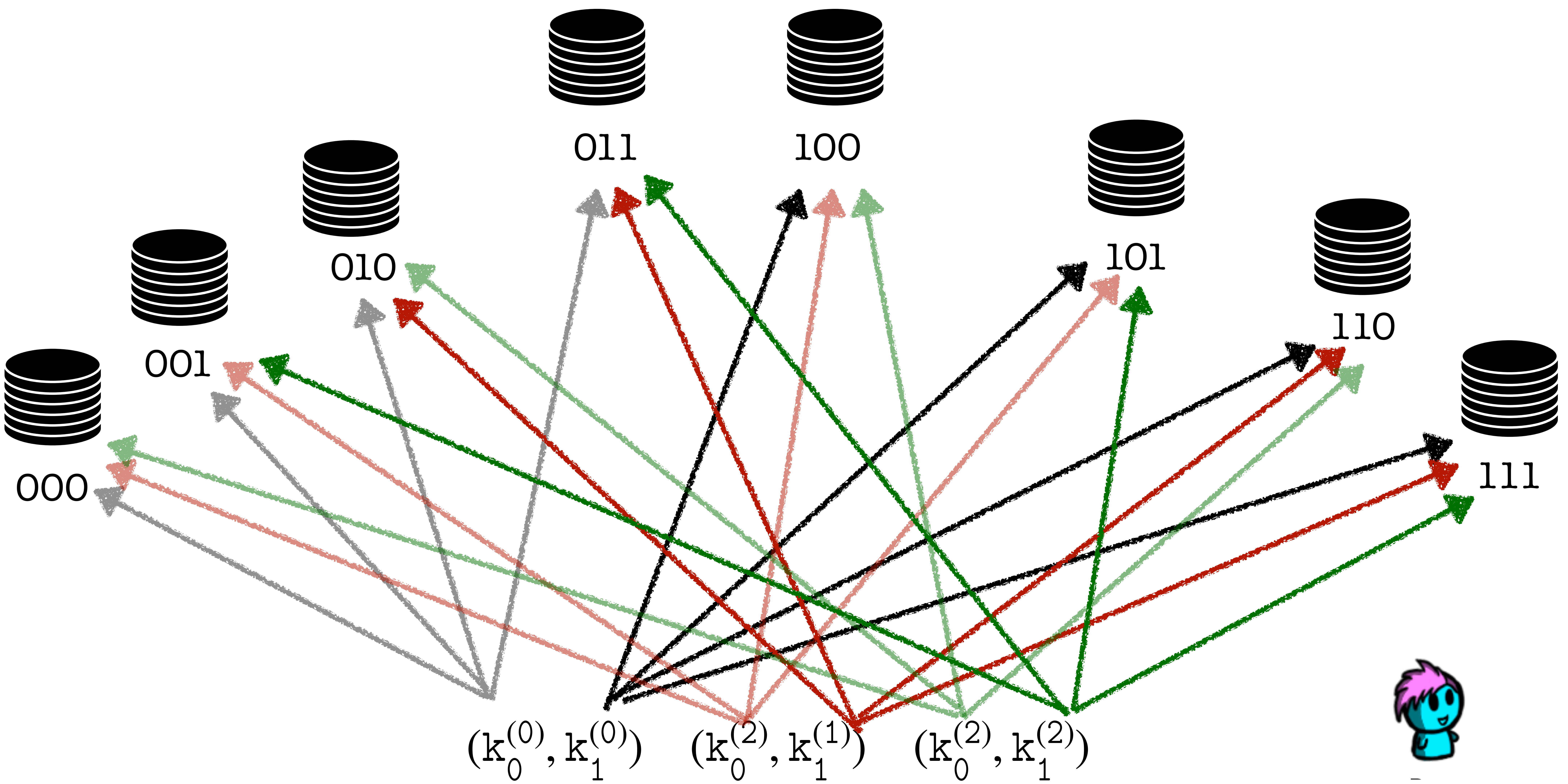
Next user selects a permutation: $\sigma\colon [0\,..\,s] \to [0\,..\,s]$

$$\mathrm{query}_k = q + \sigma(k) \cdot \vec{e}_j$$

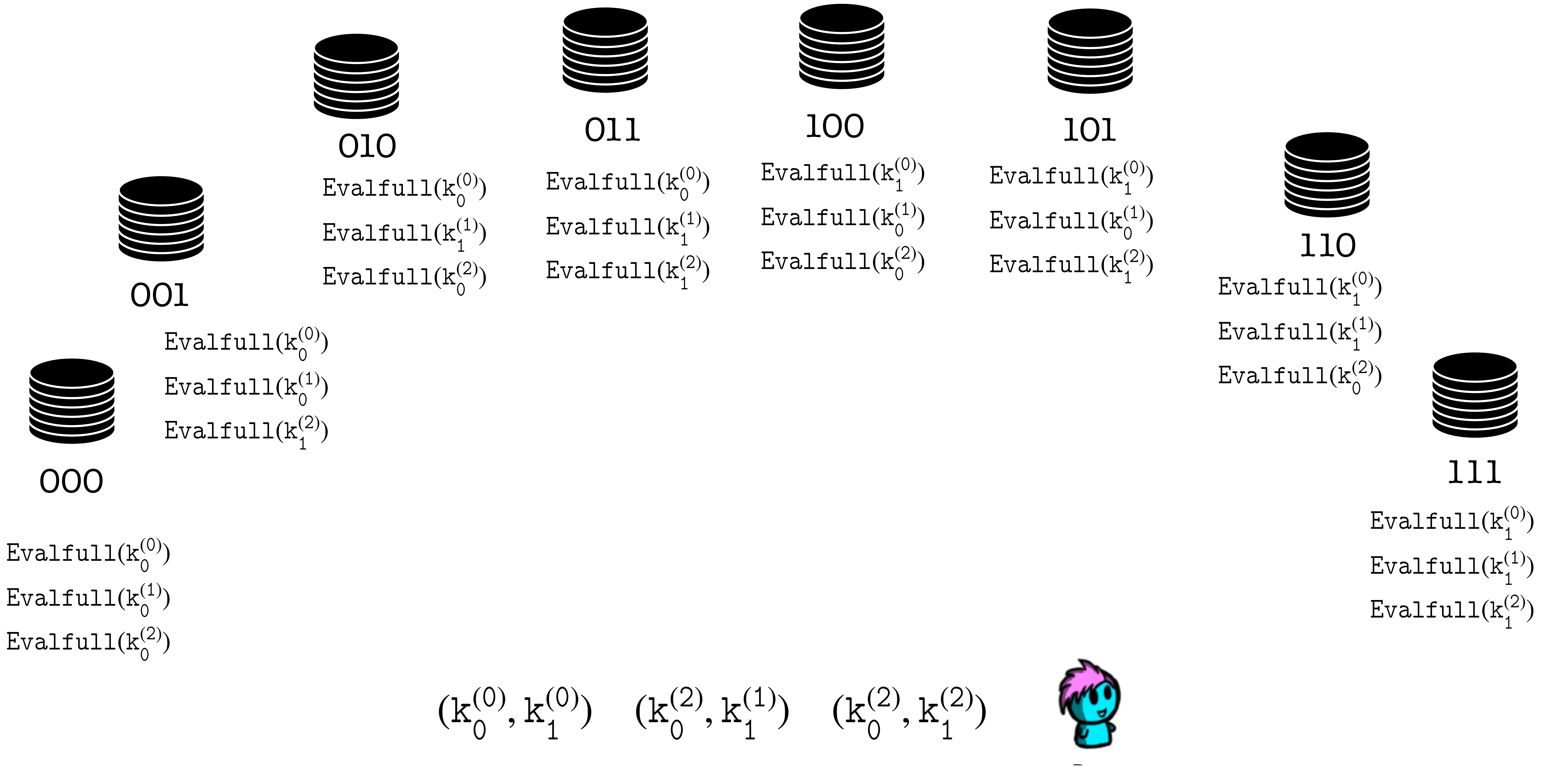
$$\mathrm{response}_k = \bigoplus_{i=1}^{r} \vec{D}_i[\ \mathrm{query}_k[i]\ ]$$

Use computes $\vec{D}_j[\sigma(k)] = \mathrm{response}_k \oplus \mathrm{respnse}_0 \quad k = 1, \ldots, s$

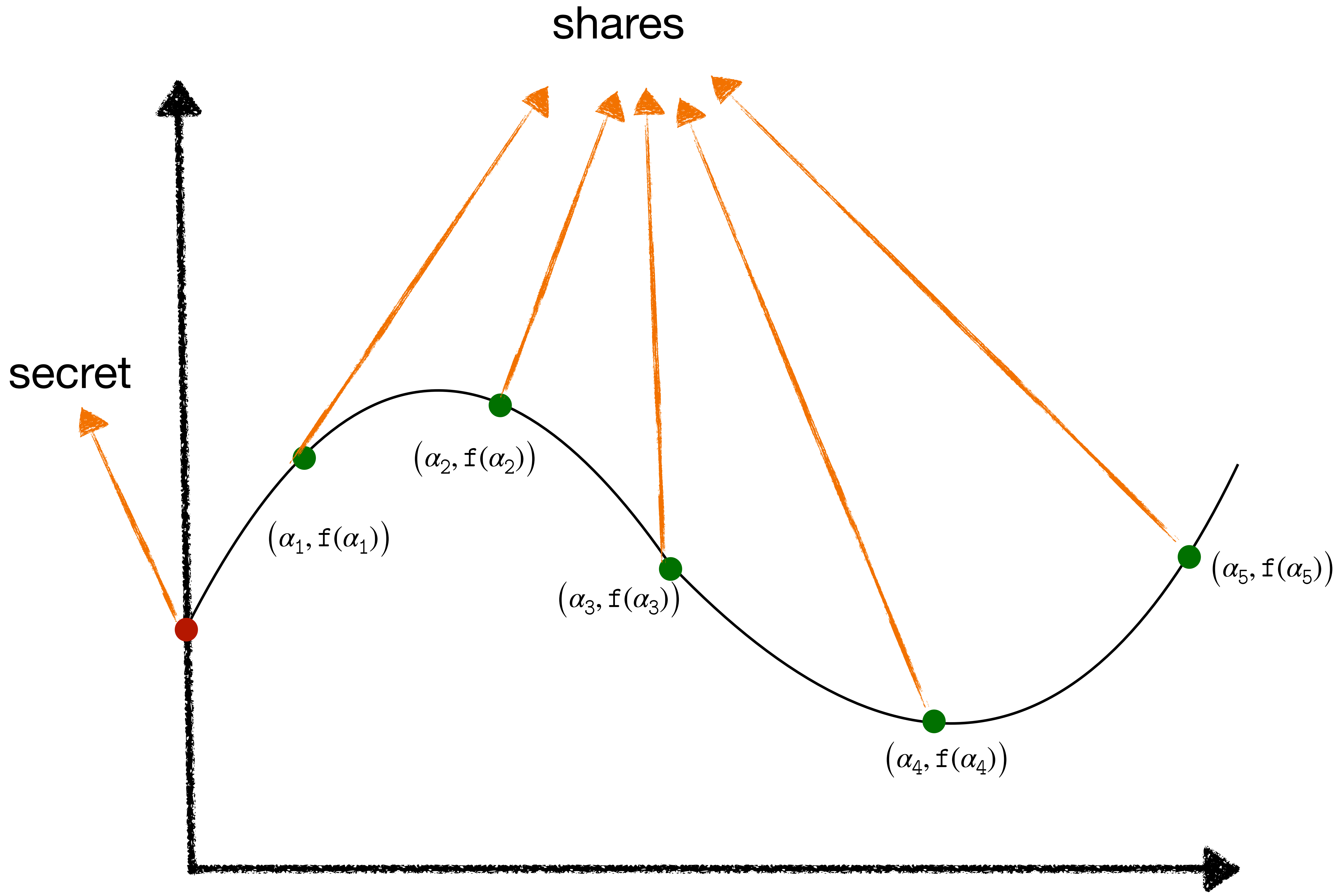
IT-Private Information Retrieval



IT-Private Information Retrieval



Shamir Secret Sharing



Properties

Linearity

$$\mathtt{f}(\mathtt{x}) = \mathtt{s}_1 + \mathtt{a}_1\mathtt{x} + \mathtt{a}_2\mathtt{x}^2 + \ldots + \mathtt{a}_t\mathtt{x}^t$$

$$\mathtt{g}(\mathtt{x}) = \mathtt{s}_2 + \mathtt{b}_1\mathtt{x} + \mathtt{b}_2\mathtt{x}^2 + \ldots + \mathtt{b}_t\mathtt{x}^t$$

$$\mathtt{h}(\mathtt{x}) = \alpha \cdot \mathtt{f}(\mathtt{x}) + \beta \cdot \mathtt{g}(\mathtt{x})$$

constant term is: $\alpha \cdot \mathtt{s}_1 + \beta \cdot \mathtt{s}_2$

each party $\mathtt{i}$ can locally compute:

$$\mathtt{h}(\mathtt{i}) = \alpha \cdot \mathtt{f}(\mathtt{i}) + \beta \cdot \mathtt{g}(\mathtt{i})$$

Shamir Secret Sharing

Properties

Linearity

Why is linearity an important property?

$$f(x) = s_1 + a_1 x + a_2 x^2 + \ldots + a_t x^t$$

$$g(x) = s_2 + b_1 x + b_2 x^2 + \ldots + b_t x^t$$

What other important property does Shamir's Secret Sharing have?

$$h(x) = \alpha \cdot f(x) + \beta \cdot g(x)$$

constant term is: $\alpha \cdot s_1 + \beta \cdot s_2$

each party i can locally compute:

$$h(i) = \alpha \cdot f(i) + \beta \cdot g(i)$$

Computational PIR using Fully Homomorphic Encryption

What is Fully Homomorphic Encryption?

$$(\texttt{pk}\ ,\ \texttt{sk}) \leftarrow \texttt{KeyGen}(1^\lambda)$$

$$\texttt{Enc}(\texttt{pk}, \texttt{x}_0) + \texttt{Enc}(\texttt{pk}, \texttt{x}_1) = \texttt{Enc}(\texttt{pk}, \texttt{x}_0 + \texttt{x}_1)$$

$$\texttt{Enc}(\texttt{pk}, \texttt{x}_0) \cdot \texttt{Enc}(\texttt{pk}, \texttt{x}_1) = \texttt{Enc}(\texttt{pk}, \texttt{x}_0 \cdot \texttt{x}_1)$$

How can FHE can be used in PIR?

Keyword PIR

What do you think is Keyword-Based PIR?

We have assumed that the client knows the exact database index of the record they are looking for.

That is not always useful.

You would rather want: $(\texttt{key}, \texttt{value})$

Two Hashes

A regular SHA256 Hash of 32 bytes

A Truncated Hash

$$(\mathtt{keyword}\ ,\ \mathtt{value}) = (\mathtt{w}, \mathtt{v})$$

$$\mathtt{V}(\mathtt{w}) = \mathtt{H}(\mathtt{w}) \| \mathtt{v}$$

Keyword based PIR

$$(\mathtt{k}_0, \mathtt{k}_1) \leftarrow \mathtt{Gen}(\mathtt{r}, \mathtt{i}, 1)$$

$$\bigoplus_{\mathtt{j}=1}^{\mathtt{r}} \mathtt{D[j]}|_{\mathtt{Eval}(\mathtt{b},\mathtt{k}_\mathtt{b},\mathtt{j})=1}$$

$$(\mathtt{k}_0, \mathtt{k}_1) \leftarrow \mathtt{Gen}(2^{\mathtt{d}}, \mathtt{H}_\mathtt{d}(\mathtt{w}), 1)$$

$$\bigoplus_{\mathtt{w}\in\mathtt{keywords}} \mathtt{V(w)}|_{\mathtt{Eval}(\mathtt{b},\mathtt{k}_\mathtt{b},\mathtt{H}_\mathtt{d}(\mathtt{w})=1)=1}$$

Theorem about achieving Robustness

Theorem: For any 1-private, (2,2)-PIR protocol, with communication cost X, one can construct a 1-private,

$(\mathtt{q}_0^{(0)}, \mathtt{q}_1^{(0)}) \leftarrow \mathtt{Query}(1^\lambda, \mathtt{j})$

$(\mathtt{q}_0^{(1)}, \mathtt{q}_1^{(1)}) \leftarrow \mathtt{Query}(1^\lambda, \mathtt{j})$

$(\mathtt{q}_0^{(\lg \ell)}, \mathtt{q}_1^{(\lg \ell)}) \leftarrow \mathtt{Query}(1^\lambda, \mathtt{j})$

$\mathtt{b}_\mathtt{i}^{(0)}\ \mathtt{b}_\mathtt{i}^{(1)}\ \mathtt{b}_\mathtt{i}^{(2)} \ldots\ \mathtt{b}_\mathtt{i}^{(\lg \ell)}$

binary representation of i

Theorem about achieving Robustness

$(\mathtt{q}_0^{(0)}, \mathtt{q}_1^{(0)}) \leftarrow \mathtt{Query}(1^\lambda, \mathtt{j})$

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binary representation of i

$\mathtt{q}_0^{(0)}, \mathtt{q}_0^{(1)}, \ldots, \mathtt{q}_0^{(\lg \ell)}$

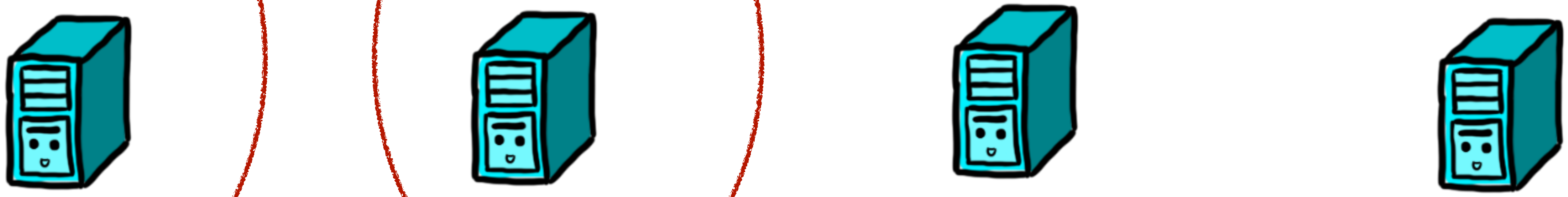
$\mathtt{q}_{\mathtt{b}_\mathtt{i}^{(0)}}^{(0)}, \mathtt{q}_{\mathtt{b}_\mathtt{i}^{(1)}}^{(1)}, \ldots, \mathtt{q}_{\mathtt{b}_\mathtt{i}^{(\lg \ell)}}^{(\lg \ell)}$

Theorem about achieving Robustness

$q_0^{(1)}, q_1^{(1)} \leftarrow \texttt{Query}(1^\lambda, j)$

$q_0^{(2)}, q_1^{(2)} \leftarrow \texttt{Query}(1^\lambda, j)$

$\left(q_0^{(1)}, q_0^{(2)}\right)$ $\left(q_0^{(1)}, q_1^{(2)}\right)$ $\left(q_1^{(1)}, q_0^{(2)}\right)$ $\left(q_1^{(1)}, q_1^{(2)}\right)$



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Theorem about achieving Robustness

$\mathsf{q}_0^{(1)}, \mathsf{q}_1^{(1)} \leftarrow \mathtt{Query}(1^\lambda, \mathsf{j})$

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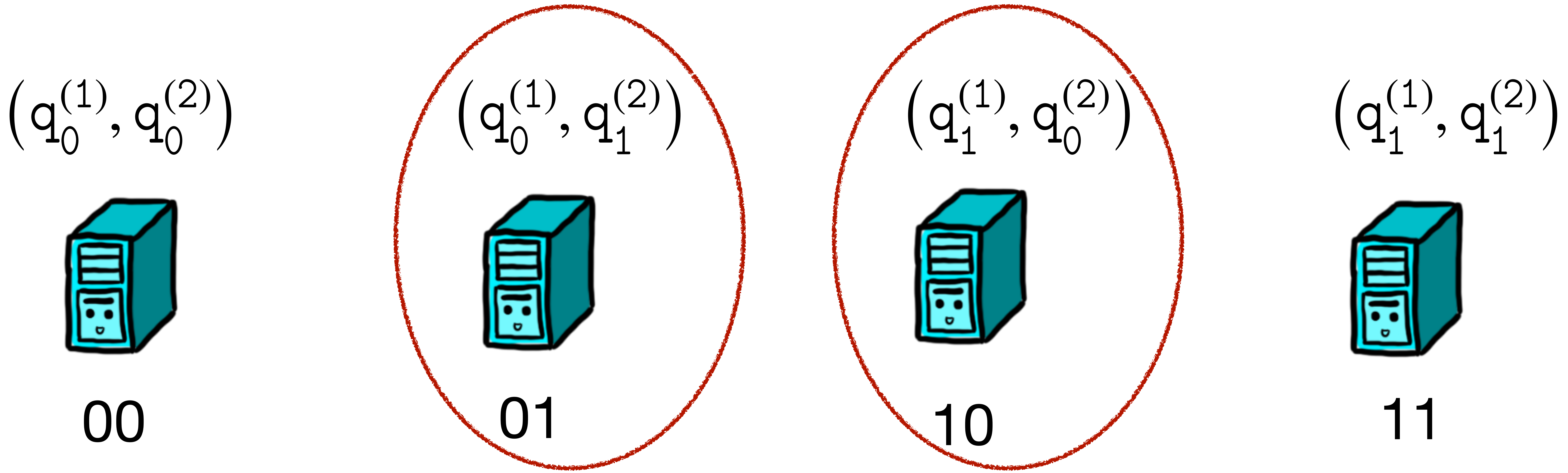

Theorem about achieving Robustness

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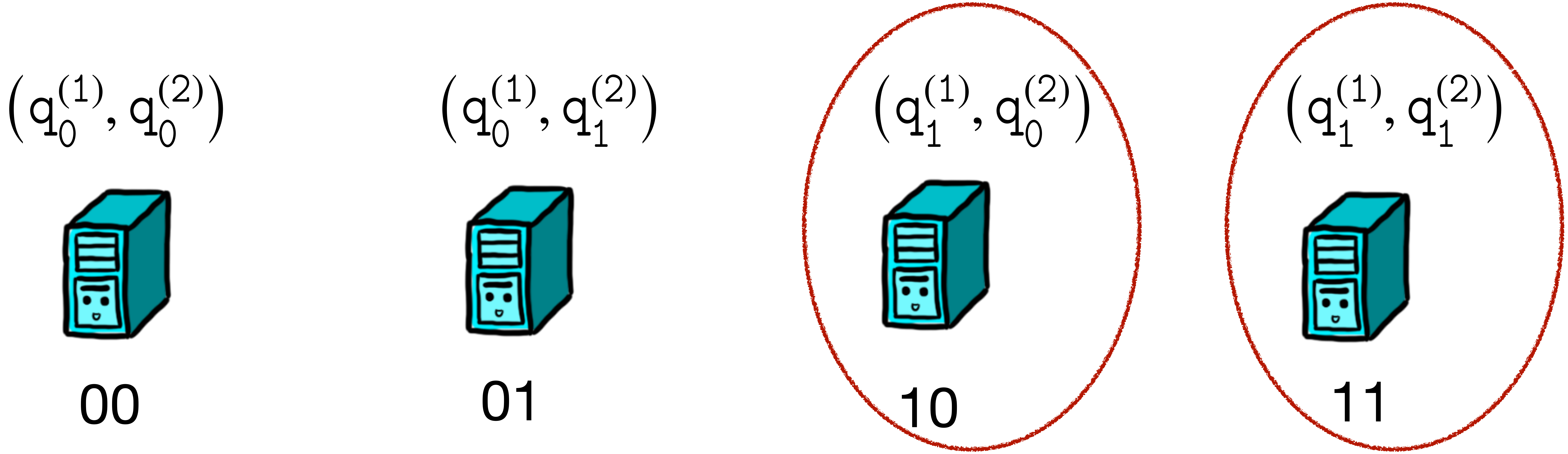

Theorem about achieving Robustness

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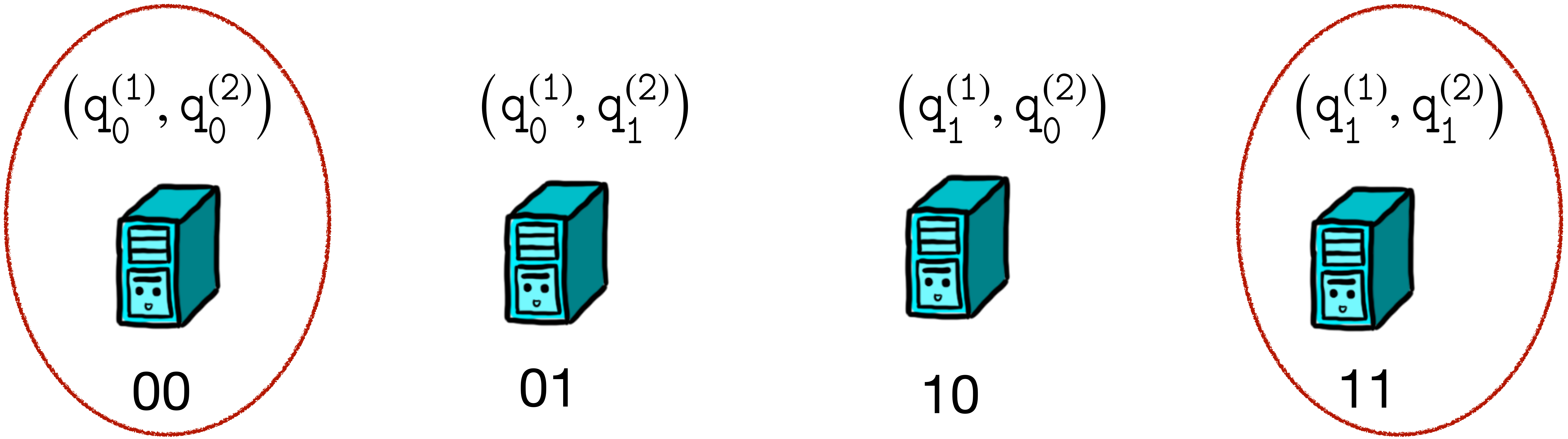
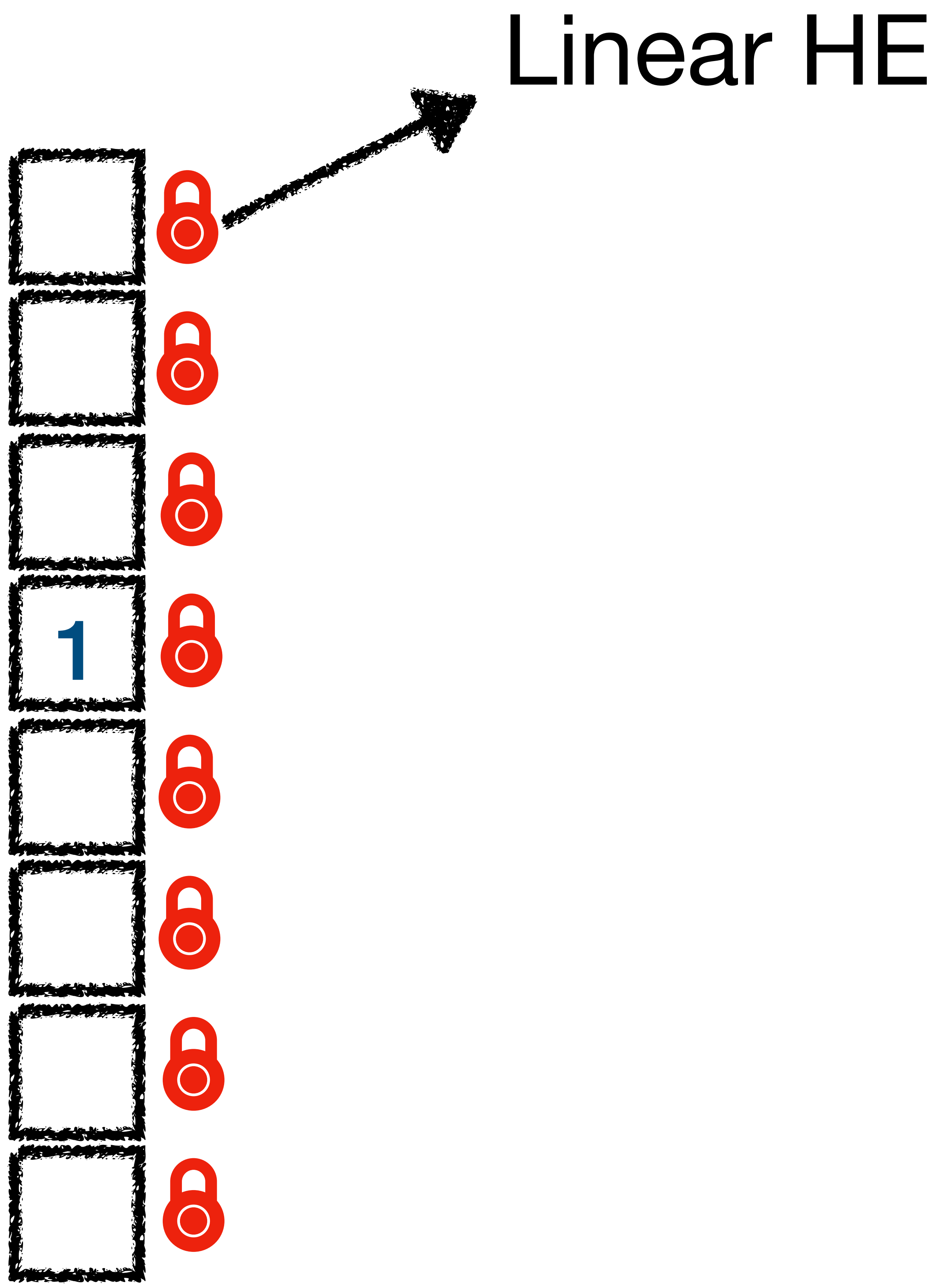
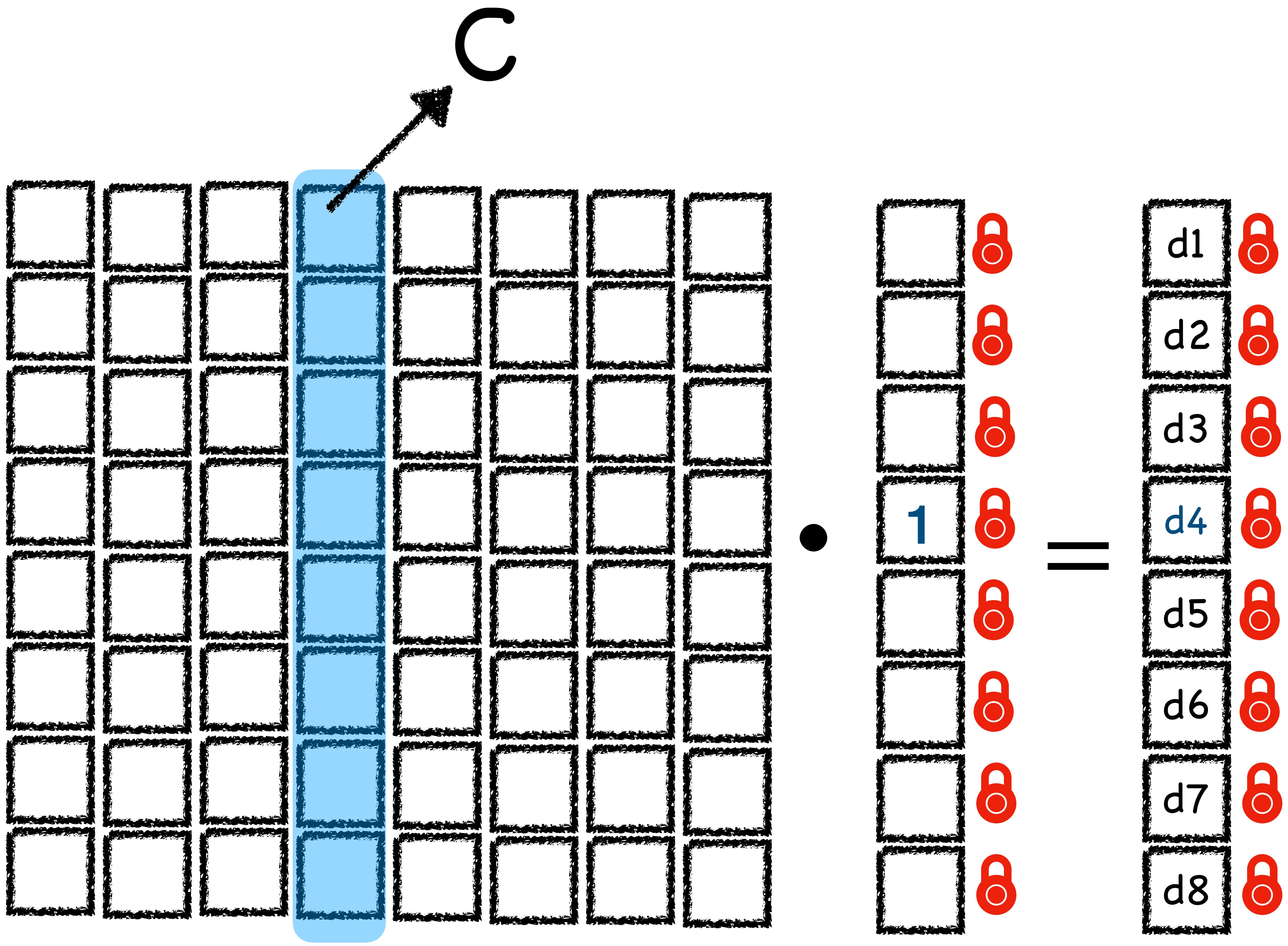


Computational PIR



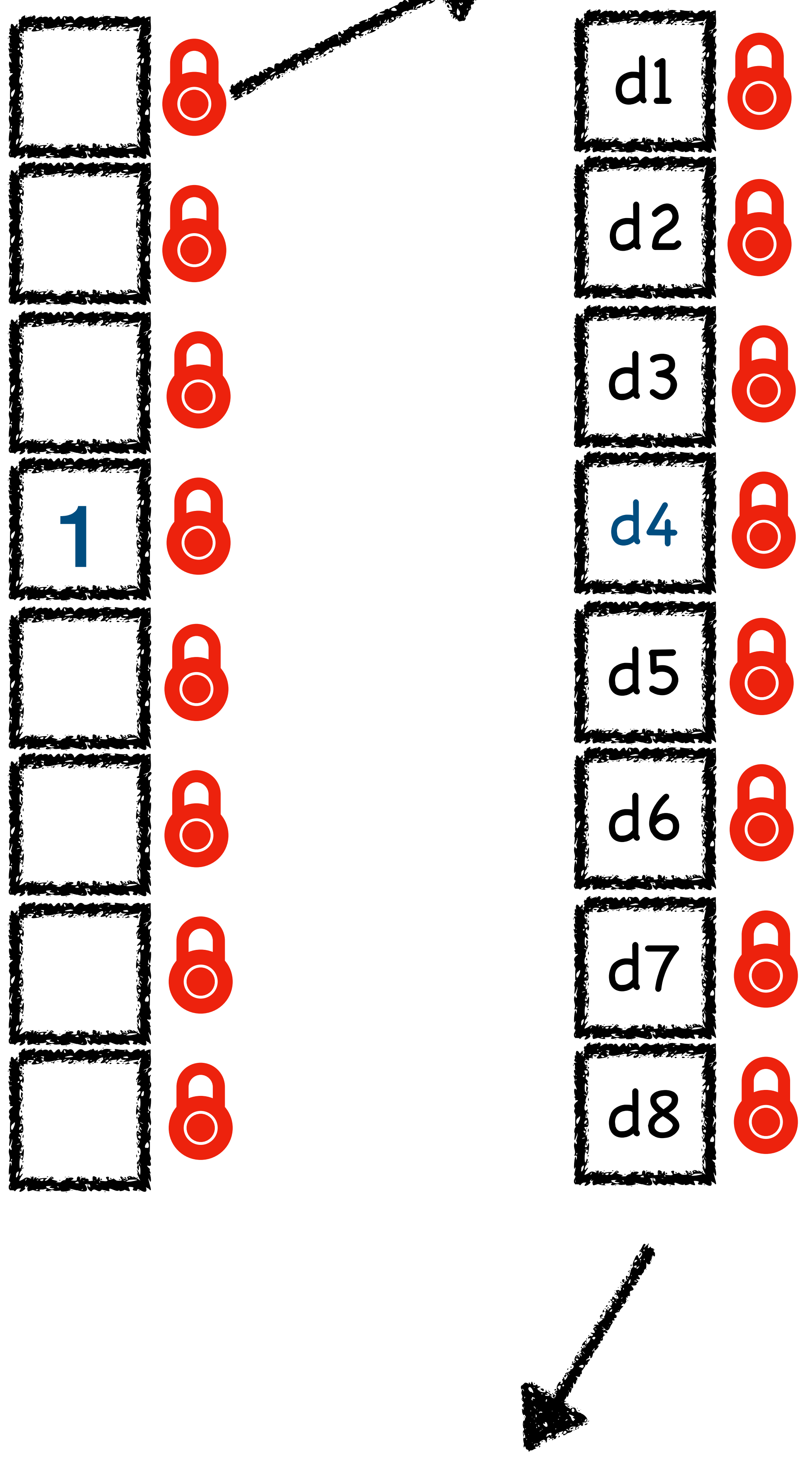


$n \times n$ matrix

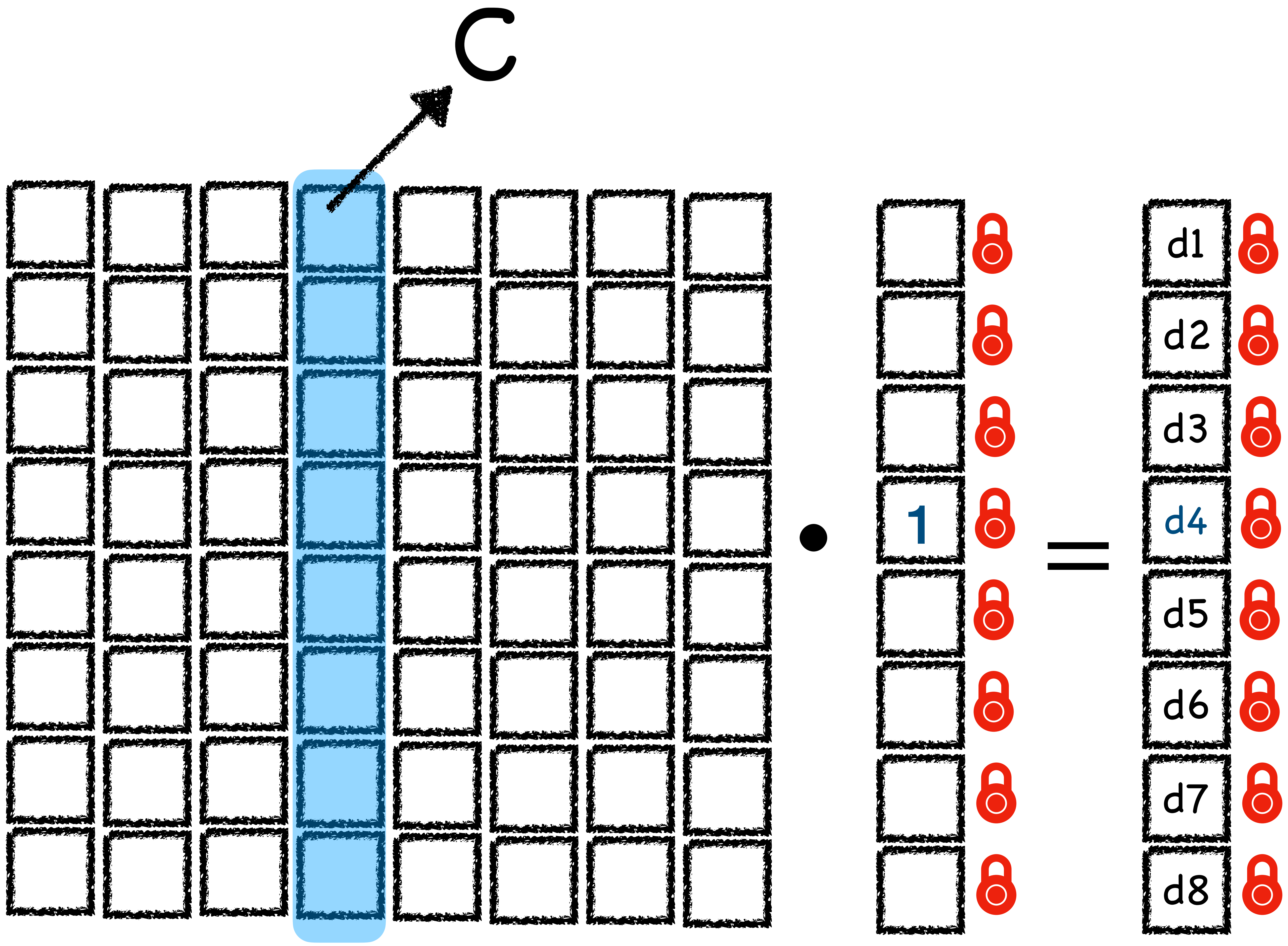
The client wants something in column C

Computational PIR

Linear HE



Decrypts the vector



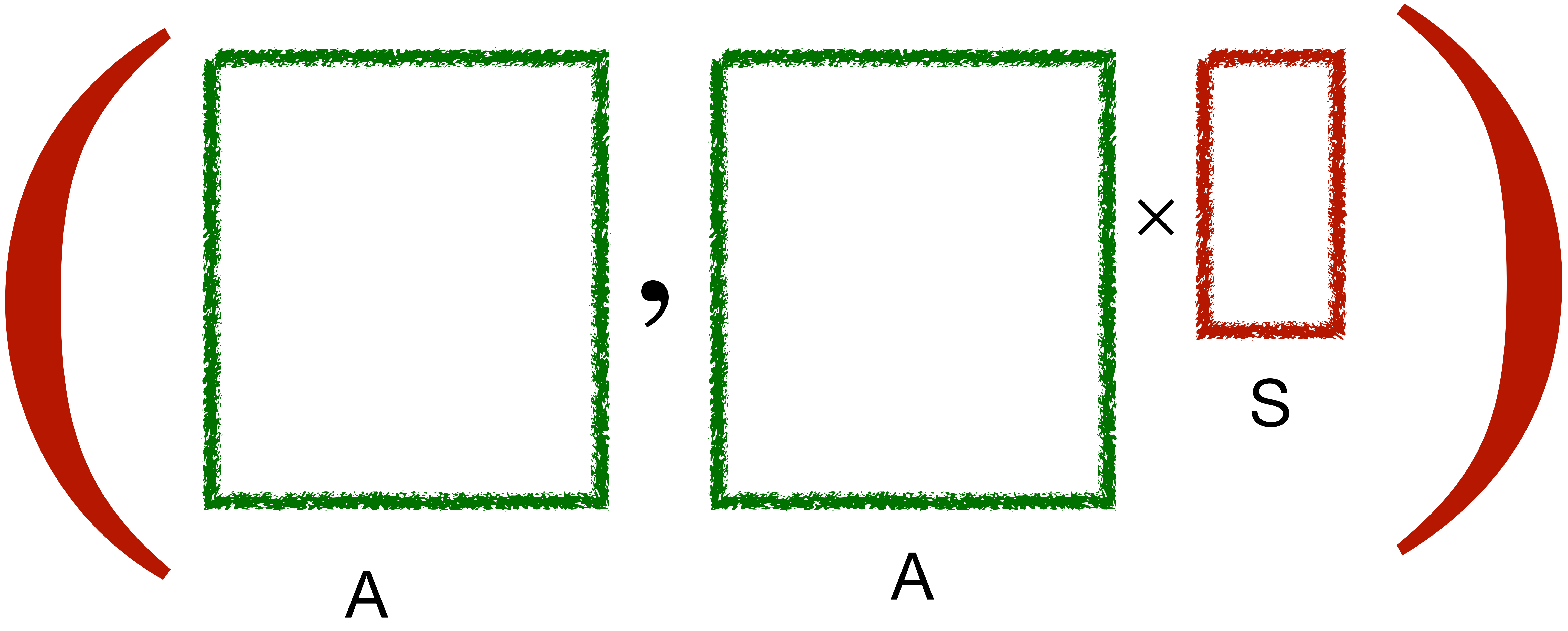
Computational PIR

Regev Encodings

Regev encoding of a scalar $\mathtt{m} \in \mathtt{R}_\mathtt{q}$

$$[-\mathtt{s} \quad 1] \cdot [\mathtt{c}_0 \quad \mathtt{c}_1]^\mathtt{T} \approx \mathtt{m}$$

Learning with Errors Problem

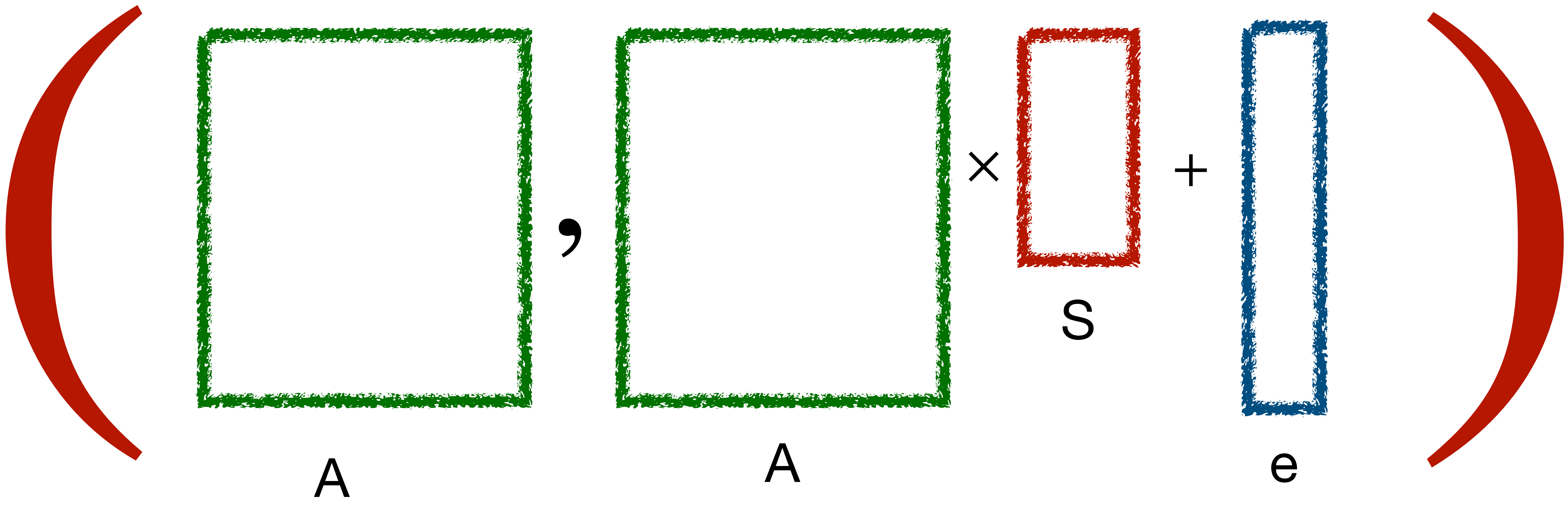


Can we find **S**?

Yes, solve the system of linear equations!

Learning with Errors Problem

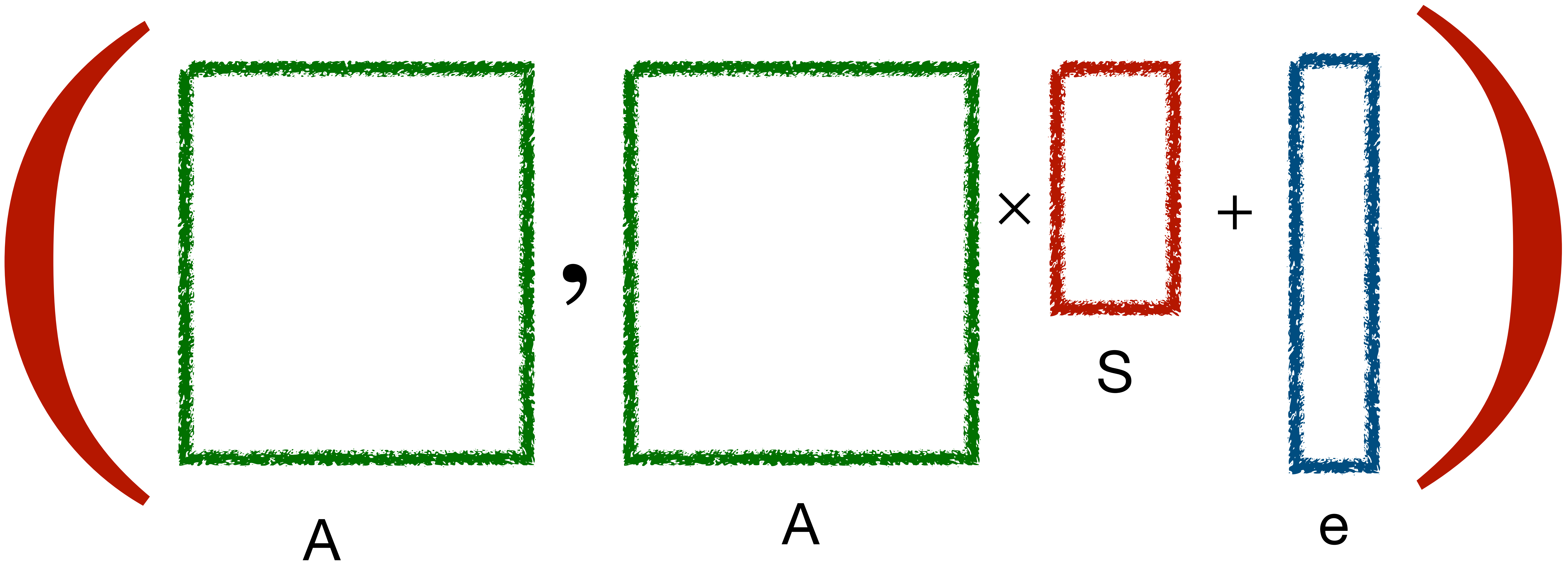
Hard to distinguish this pair from a random pair



Can we find **S**?

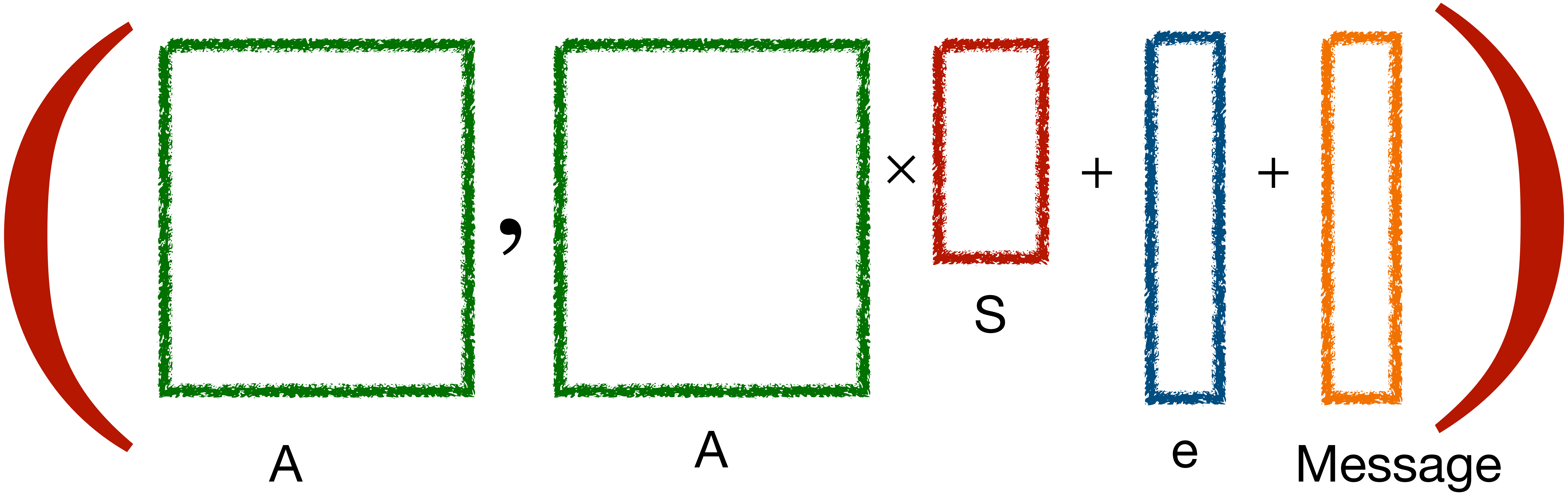
We believe the answer is NO!

Learning with Errors Problem



How can LWE be used to perform encryption?

Learning with Errors Problem



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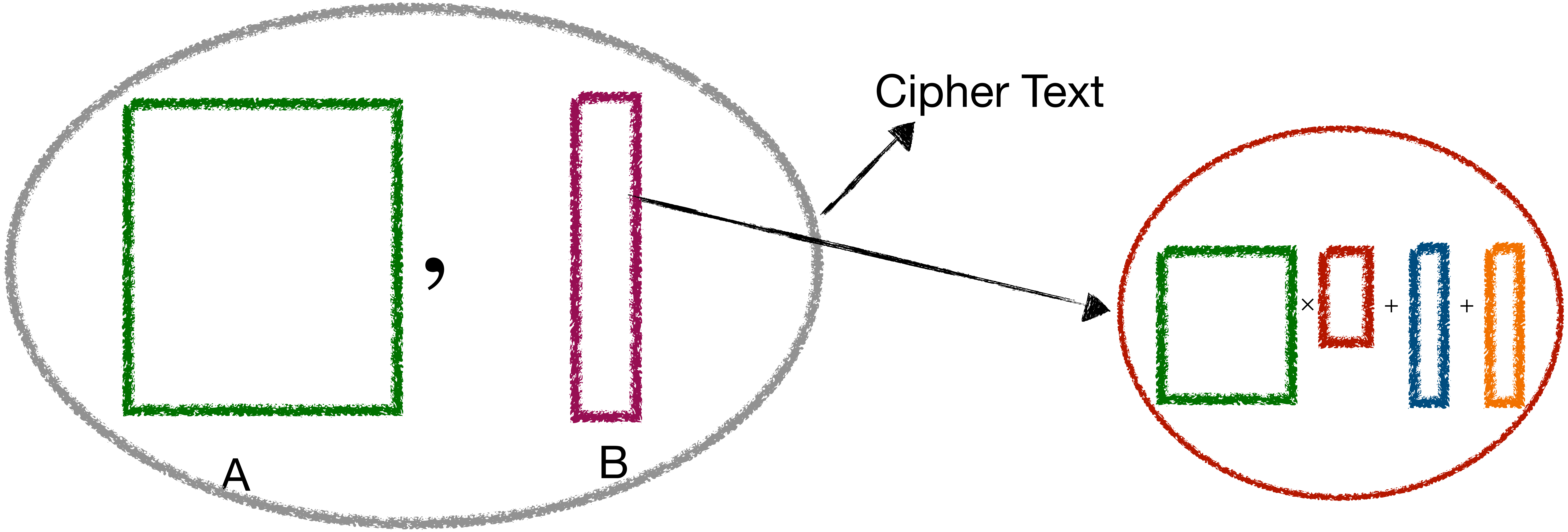

How can LWE be used to perform encryption?

B-A x S = Message (approx)

Properties of Regev Encryption

The matrix A is independent of the encrypted message

The scheme is linearly homomorphic

(A, b) Encrypts e_{i*}

$(D \cdot A, D \cdot b)$ Encrypts $D \cdot e_{i*}$

Decrypting a single element of the message requires reading a single row of $A \| b$

Learning with Errors Problem

We assume the following:

$\mathtt{q}$: be a modulus (typically a power of 2)

$\mathtt{n}$: be the dimension

$\mathtt{s} \in \mathbb{Z}_{\mathtt{q}}^{\mathtt{n}}$: be a secret vector

χ : be an error distribution over $\mathbb{Z}_{\mathtt{q}}$

$$(\vec{a}, b = \langle \vec{a}, \vec{s} \rangle + e) \mod q$$

Problem: Given many independent samples $(\vec{a}_i,\, b_i)$ distinguish them from $(\mathbb{Z}_q^n \times \mathbb{Z}_q)$

Alternatively, find the secret vector $\mathtt{S}$

Ring Learning with Errors Problem

$$\mathtt{R} = \mathbb{Z}[\mathtt{x}]/(\mathtt{f}(\mathtt{x}))$$

$$\mathbb{R}_\mathtt{q} = \mathbb{R}[/\mathtt{q}\mathbb{R} = (\mathbb{Z}_\mathtt{q}[\mathtt{x}]/(\mathtt{f}(\mathtt{x})))$$

Pick a secret $\mathtt{s} \in \mathtt{R}_\mathtt{q} \quad \mathtt{a} \in \mathtt{R}_\mathtt{q} \quad \mathtt{e} \leftarrow \chi$

$$(\mathtt{a}, \mathtt{b} = \mathtt{a} \cdot \mathtt{s} + \mathtt{e}) \mod \mathtt{q}$$

$$\mathtt{R}_\mathtt{q} \times \mathtt{R}_\mathtt{q}$$

Client-Server ORAMs

Outsources the storage to Bob

Of course, Alice will encrypt the database

But, Bob can of course, observe the access patterns!